

Activity-1

Marlon Alen I Flores
BSCS - 3A

CS311
1980-01-01

Code

```
1  using System;
2  using Avalonia.Controls;
3  using Avalonia.Interactivity;
4
5  namespace frmPrelimActivity1;
6
7  public class CleanException : System.Exception
8  {
9      public CleanException(string message) : base(message) { }
10
11     // Overriding this removes the "System.Exception:" prefix when printing the
12     // exception
13     public override string ToString()
14     {
15         return Message;
16     }
17
18     public partial class MainWindow : Window
19     {
20         public MainWindow()
21         {
22             InitializeComponent();
23         }
24
25         private bool parseOpposite(out double opposite)
26         {
27             DataValidationErrors.ClearErrors(OppositeInput);
28             if (String.IsNullOrEmpty(OppositeInput.Text))
29             {
30                 DataValidationErrors.SetError(OppositeInput,
31                 new CleanException("Opposite should not be empty."));
32                 opposite = 0;
33                 return false;
34             }
35
36             if (!double.TryParse(OppositeInput.Text, out opposite))
37             {
38                 DataValidationErrors.SetError(OppositeInput,
```

```
39         new CleanException("Opposite should be a valid number.));
40         return false;
41     }
42
43     return true;
44 }
45
46 private bool parseHypotenuse(out double hypotenuse)
47 {
48     DataValidationErrors.ClearErrors(HypotenuseInput);
49     if (String.IsNullOrEmpty(HypotenuseInput.Text))
50     {
51         DataValidationErrors.SetError(HypotenuseInput,
52             new CleanException("Hypotenuse should not be empty.));
53         hypotenuse = 0;
54         return false;
55     }
56
57     if (!double.TryParse(HypotenuseInput.Text, out hypotenuse))
58     {
59         DataValidationErrors.SetError(HypotenuseInput,
60             new CleanException("Hypotenuse should be a valid number.));
61         return false;
62     }
63
64     return true;
65 }
66
67 private bool validateAdjacent(out double adjacent)
68 {
69     DataValidationErrors.ClearErrors(AdjacentInput);
70     if (String.IsNullOrEmpty(AdjacentInput.Text))
71     {
72         DataValidationErrors.SetError(AdjacentInput,
73             new CleanException("Adjacent should not be empty.));
74         adjacent = 0;
75         return false;
76     }
77
78     if (!double.TryParse(AdjacentInput.Text, out adjacent))
79     {
80         DataValidationErrors.SetError(AdjacentInput,
81             new CleanException("Adjacent should be a valid number.));
82         return false;
83     }
84
85     return true;
```

```

86     }
87
88     private void ComputeSineButton_OnClick(object? sender, RoutedEventArgs e)
89     {
90         // singular & to disable short-circuiting
91         if (parseOpposite(out double opposite) & parseHypotenuse(out double
hypotenuse))
92         {
93             double sine = opposite / hypotenuse;
94             SineOutput.Text = sine.ToString("0.00");
95         }
96     }
97
98     private void ComputeCosineButton_OnClick(object? sender, RoutedEventArgs e)
99     {
100         if (validateAdjacent(out double adjacent) & parseHypotenuse(out double
hypotenuse))
101         {
102             double cosine = adjacent / hypotenuse;
103             CosineOutput.Text = cosine.ToString("0.00");
104         }
105     }
106
107     private void ComputeTangentButton_OnClick(object? sender, RoutedEventArgs
e)
108     {
109         if (parseOpposite(out double opposite) & validateAdjacent(out double
adjacent))
110         {
111             double tangent = opposite / adjacent;
112             TangentOutput.Text = tangent.ToString("0.00");
113         }
114     }
115
116     private void ClearAllButton_OnClick(object? sender, RoutedEventArgs e)
117     {
118         DataValidationErrors.ClearErrors(OppositeInput);
119         DataValidationErrors.ClearErrors(HypotenuseInput);
120         DataValidationErrors.ClearErrors(AdjacentInput);
121         OppositeInput.Text = "";
122         HypotenuseInput.Text = "";
123         AdjacentInput.Text = "";
124         SineOutput.Text = "";
125         CosineOutput.Text = "";
126         TangentOutput.Text = "";
127     }
128

```

```

129     private void ClearButton_OnClick(object? sender, RoutedEventArgs e)
130     {
131         OppositeInput.Text = "";
132         HypotenuseInput.Text = "";
133         AdjacentInput.Text = "";
134         DataValidationErrors.ClearErrors(OppositeInput);
135         DataValidationErrors.ClearErrors(HypotenuseInput);
136         DataValidationErrors.ClearErrors(AdjacentInput);
137         SineOutput.Text = "";
138         CosineOutput.Text = "";
139         TangentOutput.Text = "";
140     }
141
142     private void ComputeAllButton_OnClick(object? sender, RoutedEventArgs e)
143     {
144         ComputeSineButton_OnClick(sender, e);
145         ComputeCosineButton_OnClick(sender, e);
146         ComputeTangentButton_OnClick(sender, e);
147     }
148 }

```

Screenshots

1. Opposite and Hypotenuse are empty and click sine button.

Adjacent

Hypotenuse
Hypotenuse should not be empty.

Opposite
Opposite should not be empty.

Compute Sine Compute Cosine Compute Tangent

Compute All Clear All

Sine

Cosine

Tangent

2. Adjacent and Hypotenuse are empty and click cosine button.

Adjacent

Adjacent should not be empty.

Hypotenuse

Hypotenuse should not be empty.

Opposite

Sine

Cosine

Tangent

3. Opposite and Adjacent are empty and click tangent button.

Adjacent

Adjacent should not be empty.

Hypotenuse

Opposite

Opposite should not be empty.

Sine

Cosine

Tangent

4. All inputs are empty and click compute all button.

Adjacent

Adjacent should not be empty.

Hypotenuse

Hypotenuse should not be empty.

Opposite

Opposite should not be empty.

Sine

Cosine

Tangent

5. Adjacent is not a number.

Adjacent

Adjacent should be a valid number.

Hypotenuse

Hypotenuse should not be empty.

Opposite

Sine

Cosine

Tangent

6. Hypotenuse is not a number.

Adjacent

Hypotenuse
Hypotenuse should be a valid number.

Opposite
Opposite should not be empty.

Sine

Cosine

Tangent

7. Opposite is not a number.

Adjacent
Adjacent should not be empty.

Hypotenuse

Opposite
Opposite should be a valid number.

Sine

Cosine

Tangent

8. Hypotenuse is 3 and Opposite is 6 and click sine button.

Adjacent	
Hypotenuse	3
Opposite	6
Compute <u>S</u>ine Compute C<u>o</u>sine Compute <u>T</u>angent	
Compute <u>A</u>ll <u>C</u>lear All	
Sine	2.00
Cosine	
Tangent	

9. Hypotenuse is 6 and Opposite is 3 and click sine button. Result should be in 2 decimal places only.

Adjacent	
Hypotenuse	6
Opposite	3
Compute <u>S</u>ine Compute C<u>o</u>sine Compute <u>T</u>angent	
Compute <u>A</u>ll <u>C</u>lear All	
Sine	0.50
Cosine	
Tangent	

10. Adjacent is 10 and Hypotenuse is 5 and click cosine button.

Adjacent

Hypotenuse

Opposite

Sine

Cosine

Tangent

11. Adjacent is 5 and Hypotenuse is 10 and click cosine button. Result should be in 2 decimal places only.

Adjacent

Hypotenuse

Opposite

Sine

Cosine

Tangent

12. Adjacent is 8 and Opposite is 5 and click tangent button. Result should be in 2 decimal places only.

Adjacent	8
Hypotenuse	
Opposite	5
Compute <u>S</u>ine Compute <u>C</u>osine Compute <u>T</u>angent	
Compute <u>A</u>ll <u>C</u>lear All	
Sine	
Cosine	0.50
Tangent	0.63

13. Adjacent is 8, Hypotenuse is 5, and Opposite is 3 and click compute all button.

Adjacent	8
Hypotenuse	5
Opposite	3
Compute <u>S</u>ine Compute <u>C</u>osine Compute <u>T</u>angent	
Compute <u>A</u>ll <u>C</u>lear All	
Sine	0.60
Cosine	1.60
Tangent	0.38

14. Adjacent is 3, Hypotenuse is 5, and Opposite is 8 and click compute all button.

Adjacent	3
Hypotenuse	5
Opposite	4
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	1.60
Cosine	0.60
Tangent	2.67

15. Adjacent is 5, Hypotenuse is 8, and Opposite is 3 and click compute all button. Result should be in 2 decimal places only

Adjacent	5
Hypotenuse	8
Opposite	3
Compute Sine Compute Cosine Compute Tangent	
Compute All Clear All	
Sine	0.38
Cosine	0.63
Tangent	0.60

https://drive.google.com/file/d/1Um74ZM7_IOZKvRXvlseefelyWHuBKqe7/view?usp=sharing